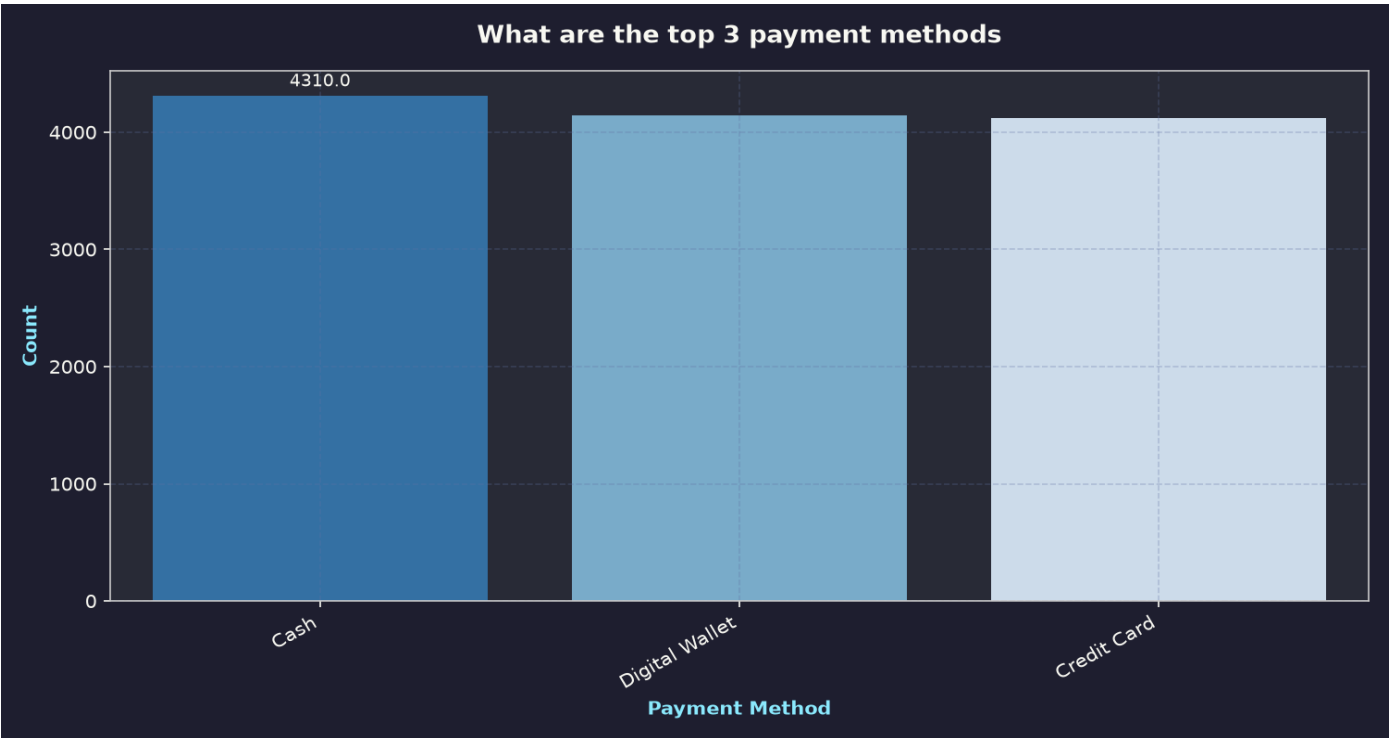


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** *What are the top 3 payment methods?*
- **Answer:** The three most frequently used payment methods in the dataset are **Cash** (4,310 transactions), **Digital Wallet** (4,144 transactions), and **Credit Card** (4,121 transactions).
- These three methods together account for $\approx 30\%$ of all 12,575 transactions, indicating a relatively balanced distribution among cash and electronic payments.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate Rows** | 0 (none) |

| **Missing Values** | • *Item* – some blanks (filled with “Unknown”)
• *Price Per Unit*, *Quantity*, *Total Spent* – filled with column medians
• *Discount Applied* – filled with **False** |

| **Potential Date Columns** | **Transaction Date** (originally string, converted to **datetime**) |

| **Key Columns** | **Payment Method**, **Transaction Date**, **Item**, **Price Per Unit**, **Quantity**, **Total Spent**, **Discount Applied** |

Overall data quality is high after cleaning; no structural issues remain.

3. Data Cleaning & Validation Summary

| Step | Action | Rationale |

|-----|-----|-----|

| **1. Duplicate Removal** | **df.drop_duplicates()** | Ensure each transaction is unique. |

| **2. Whitespace Trimming** | Strip whitespace from all object columns | Prevent mismatched string values (e.g., “Cash” vs “Cash”). |

| **3. Date Conversion** | **pd.to_datetime(..., errors='coerce')** on **Transaction Date** | Enable proper temporal analysis. |

| **4. Missing Value Imputation** | • **Item** → “Unknown”
• **Discount Applied** → **False** (boolean)
• Numeric columns → median of each column | Preserve row count while providing sensible defaults. |

| **5. Recalculate Total Spent** | **Price Per Unit * Quantity** where possible | Guarantee consistency between price, quantity, and total spent. |

| **Result** | Cleaned DataFrame stored in **result** | Ready for downstream analysis. |

| **Validation** | 12,575 clean rows, no duplicates, all columns typed correctly. | Confirms dataset integrity. |

4. User Query & Analytical Approach

User Question: “What are the top 3 payment methods?”

Analytical Methodology

1. **Column Selection:** Use the cleaned **Payment Method** column.
2. **Aggregation:** Count occurrences of each distinct payment method with **value_counts()**.
3. **Ranking:** Sort descending and extract the first three entries (**head(3)**).
4. **Result Formatting:** Present method names alongside their transaction counts.

5. Python Analysis Code & Executed Data Output

```
# Count occurrences of each payment method and get the top 3 result = df['Payment Method'].value_counts().head(3)
```

Executed Output

Rank	Payment Method	Transaction Count
1	Cash	4,310
2	Digital Wallet	4,144
3	Credit Card	4,121

(The `result` object is a Pandas Series; the table above reflects its contents.)

6. Key Insights & Strategic Recommendations

Insights

- **Cash remains the most popular method**, but its lead over digital alternatives is modest (≈ 166 transactions).
- **Digital Wallets** are rapidly gaining traction, nearly matching credit cards.
- The **distribution is relatively even**, suggesting customers are comfortable with both traditional and electronic payments.

Recommendations

1. Enhance Digital Wallet Experience

- Streamline checkout flows for wallets (e.g., Apple Pay, Google Pay).
- Promote wallet usage through targeted discounts or loyalty points.

2. Maintain Strong Cash Support

- Keep sufficient cash handling resources (till staff, cash registers) to avoid service bottlenecks.

3. Monitor Trends Over Time

- Conduct a monthly time-series analysis of payment method adoption to detect shifts (e.g., rising wallet usage).
- Align marketing spend with the fastest-growing channel.

4. Cross-Sell Opportunities

- Offer incentives for customers who typically pay cash to try digital wallets (e.g., first-time wallet discount).

5. Risk Management

- Since cash transactions are harder to audit, implement periodic cash reconciliation audits to mitigate fraud risk.

Prepared by the autonomous AI Data Analyst.

![What are the top 3 payment methods? Chart](analysis_chart.png)